

True/False Questions (1 marks * xx Questions)

Question 1

(T/F Q2) A simple reflex agent selects actions based solely on the current percept and does not consider goals or plan future actions.

- (a) True
- (b) False

Ans: (a)

MCQ Questions (1.5 marks * xx Questions)

Question 1

(MCQ Q1) Which of the following best describes the relationship between Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and Generative AI?

- (a) Machine Learning is a broader field that includes AI, Deep Learning, and Generative AI.
- (b) Deep Learning, Machine Learning, and Generative AI are three separate approaches that are unrelated to AI.
- (c) Artificial Intelligence is a subset of Deep Learning, which itself is a subset of Machine Learning, and Generative AI is unrelated.
- (d) Deep Learning is a subset of Machine Learning, which is a subset of Artificial Intelligence, and Generative AI is a specialized application of AI and ML, often using deep learning models.

Ans: (d)

Question 2

(MCQ Q2) What is the major advantage of graph search compared to tree search?

- (a) Graph search is visually easier to represent than tree search.
- (b) Graph search avoids exploring repeated states by keeping track of visited ones.
- (c) Graph search requires less memory because it stores fewer states.
- (d) There is no advantage since graph search is just another name for tree search.

Ans: (b)

Question 3

(MCQ Q3) Which of the following best describes the difference between uninformed and informed search strategies?

- (a) Uninformed search expands nodes without using problem-specific knowledge, while informed search leverages heuristics to guide the search toward the goal.
- (b) Uninformed search can guarantee optimality if the branching factor is finite, while informed search cannot guarantee optimality even with admissible heuristics.
- (c) Informed search strategies, such as A*, explore all nodes equally but stop earlier when the goal is reached, unlike uninformed strategies.
- (d) Both uninformed and informed search require heuristics, but informed search additionally requires admissibility and consistency to ensure completeness.

Ans: (a)

Question 4

(MCQ Q4) Formulate a linear program for the following scenario:

Blue Sky Solutions is a company specializing in solar panel installations. They have a limited workforce and want to maximize their profit while adhering to certain constraints. The company has two types of solar panel installations: Type A and Type B.

Each Type A installation:

- Requires 11 hours of labor
- Costs \$600 in materials
- Generates \$900 profit per installation.

Each Type B installation:

- Requires 7 hours of labor
- Costs \$400 in materials
- Generates \$700 profit per installation.

The company has a total of 1100 labor hours and \$420,000 in cash available. Due to supply constraints, the company can only install a maximum of 250 Type A panels.

Let decision variable A denote the number of Type A installations and B denote the number of Type B installations.

Which of the following is the correct formulation of the above LP problem? (Min and Max are abbreviations for Minimize and Maximize, respectively.)

- (a) $\text{Min } 11A + 7B$

$$\begin{aligned}
\text{s.t.: } & 11A + 7B \leq 1100 \\
& 600A + 400B \leq 420000 \\
& A \leq 250 \\
& A, B \geq 0
\end{aligned}$$

(b) Max $900B + 700A$

$$\begin{aligned}
\text{s.t.: } & 11A + 7B \leq 1100 \\
& 600A + 400B \leq 420000 \\
& A \leq 250
\end{aligned}$$

(c) Max $900A + 700B$

$$\begin{aligned}
\text{s.t.: } & 11A + 7B \leq 1100 \\
& 600A + 400B \leq 420000 \\
& A \leq 250 \\
& A, B \geq 0
\end{aligned}$$

(d) Max $A + B$

$$\begin{aligned}
\text{s.t.: } & 11A + 7B \leq 1100 \\
& 600A + 400B \leq 420000 \\
& A \leq 250 \\
& A, B \geq 0
\end{aligned}$$

Ans: (c)

Question 5

(MCQ Q5) The given linear program is:

Maximize $z = 3x_1 + 2x_2$
subject to:

$$\begin{aligned}
x_1 + x_2 &\leq 2 \\
4x_1 + 3x_2 &\geq 15 \\
x_1, x_2 &\geq 0
\end{aligned}$$

- (a) Bounded and feasible
- (b) Feasible but not bounded
- (c) Bounded but not feasible

(d) Neither bounded nor feasible

Ans: (d)

MRQ Questions (2 marks * xx Questions)

Question 1

(MRQ Q1) Which of the following statements about rational agents are correct? Select all that apply.

- (a) A rational agent always knows the outcome of its actions in advance.
- (b) A rational agent chooses actions that are expected to yield the highest performance according to a predefined measure.
- (c) Rational agents may not always produce the best solution in every scenario.
- (d) Smart home devices, such as automatic lights, can be considered agents.

Ans: (b) (c) (d)

Question 2

(MRQ Q2) Which of the following are reasons to use Breadth-First Search (BFS) over Depth-First Search (DFS)? Select all that apply.

- (a) BFS always finds the shortest path in an unweighted graph.
- (b) BFS guarantees completeness in finite search spaces.
- (c) BFS usually uses less memory than DFS in very deep trees.
- (d) BFS goes level by level, making it suitable for finding the shallowest solution first.
- (e) BFS is more effective than DFS for exploring very deep or infinite search spaces.

Ans: (a) (b)(d)

Question 3

(MRQ Q3) Which of the following statements about informed search algorithms are true? Select all that apply.

- (a) The efficiency of informed search algorithms is completely independent of the heuristic function used.
- (b) A better heuristic can reduce the number of nodes explored during the search.
- (c) Informed search algorithms always find the optimal solution, even if the heuristic overestimates the true cost.

(d) If $h(n)$ is a consistent heuristic and all step costs are positive, then the total estimated cost $f(n) = g(n) + h(n)$ never decreases along any path, where:

- $g(n)$ is the cost to reach node n from the start node, and
- $h(n)$ is the estimated cost from node n to the goal.

Ans: (b) (d)

Question 4

(MRQ Q4) Which of the following statements about A* search are true? Select all that apply.

- (a) A* search combines the cost from the start node to the current node and a heuristic estimate of the remaining cost.
- (b) If the heuristic is admissible, A* tree search will always return an optimal solution (assuming non-negative step costs).
- (c) A* with graph search remains optimal when the heuristic is consistent, because consistency ensures non-decreasing $f(n)$ values along a path.
- (d) A* may expand fewer nodes than Uniform Cost Search if the heuristic provides useful guidance toward the goal.
- (e) A* guarantees optimality even if the heuristic overestimates the true cost to reach the goal.

Ans: (a) (b) (c) (d)

Question 5

Which of the following functions are convex? Select all that apply.

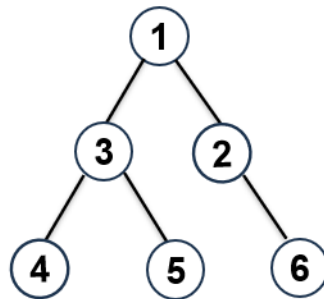
- (a) $f(x) = -x^2 + 2x + 9, x \in \mathcal{R}$
- (b) $f(x) = 2x - 3, x \in \mathcal{R}$
- (c) $f(x) = |-2x + 4|, x \in \mathcal{R}$
- (d) $f(x) = -2x^6, x \in \mathcal{R}$
- (e) $f(x) = \sqrt{x}, x \in (0, +\infty)$

Ans: (b)(c)

Fill in the Blank (In total xx questions with xx blanks, 2.5 marks for each blank, in total 2.5*xx = xx marks)

Question 1 [2 Blanks]

(FITB Q1) Consider the graph below. Start from node 1, and assume that when visiting neighbors, nodes are explored in increasing numerical order (e.g, node 2 is visited before node 3). Write down the traversal order using Breadth-First Search (BFS) and Depth-First Search (DFS), respectively.



- Enter only the numeric values representing the nodes, without any spaces or separators.
- Graph search strategy is used for both BFS and DFS (nodes are not revisited).
- A node will not be added to the frontier more than once.

Examples:

- If a traversal sequence is $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$, write: 12345
- If a traversal sequence is $1 \rightarrow 2 \rightarrow 5 \rightarrow 4 \rightarrow 3$, write: 12543

Your answers:

- (a) The traversal sequence by BFS is _____
- (b) The traversal sequence by DFS is _____

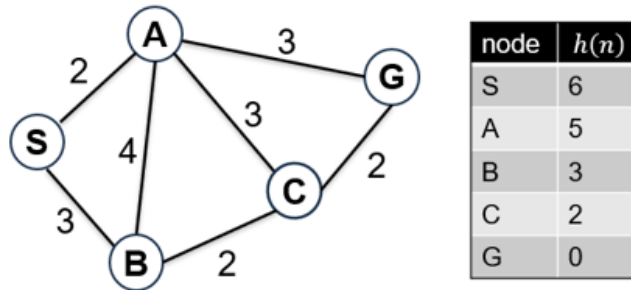
Ans:

- (a) 123645 (acceptable range: 123644.5 – 123645.5)
- (b) 126345 (acceptable range: 126344.5 – 126345.5)

Question 2 [1 Blank]

(FITB Q2) Please perform greedy best-first search on the graph below, using a graph search strategy (i.e., nodes are not revisited once expanded). The numbers on the edges represent the step cost between nodes, and the heuristic values (estimated cost to the goal G) for each node are shown in the table on the right side of the graph.

Starting from node S, determine total cost of the path found by greedy best-first search to reach the goal node G.



Answer instructions:

- Enter only digits (0–9) and an optional leading minus sign (–) if the answer is negative. Round or report your answer as an integer. Do not include letters, commas, spaces, or other characters.

Your answers:

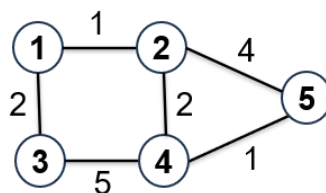
Final total path cost by greedy best-first search: BLANK1

Ans:

7 (acceptable range: 6.5 – 7.5)

Question 3 [1 Blank]

(FITB Q3) Consider the following graph. The start node is **1**, and the goal node is **5**. Using the priority queue to implement the frontier, list the order in which nodes are popped from the priority queue during Dijkstra's algorithm, until the goal node **5** is reached.



- Enter only the numeric values representing the nodes, without any spaces or separators.**
- Graph search strategy is used (nodes are not revisited).
- A node will not be added to the frontier more than once.
- The goal test is performed only when a node is removed from the frontier.
- Not every node in the graph will necessarily be popped. List only the nodes that are actually removed from the priority queue until the goal node is reached.

Examples:

- If the popped order is $1 \rightarrow 2 \rightarrow 4 \rightarrow 5$, write: 1245

Your answers:

Order of nodes popped by Dijkstra's algorithm: BLANK1

Ans:

12345 (acceptable range: 12344.5 – 12345.5)

Question 4 [1 Blank]

The given linear program is:

$$\text{Maximize } z = 7x_1 + 2x_2 + 6x_3$$

subject to:

$$2x_1 + x_2 + 3x_3 \leq 14$$

$$-x_1 + 4x_2 + x_3 \geq 3$$

$$3x_1 - 2x_2 + 2x_3 \leq 12$$

$$x_1 + 5x_2 + 2x_3 \leq 20$$

$$x_1 + x_2 + 3x_3 \geq 5$$

$$x_1, x_2, x_3 \geq 0$$

Solve this linear program. Provide the maximum objective value. Round your answer to 2 decimal places.

Your answer:

Maximum objective value (Round your answer to 2 decimal places): BLANK1

Ans:

45.14 (acceptable range: 45.04 – 45.24)

Question 5 [2 Blanks]

Suppose we are minimizing $f(x) = (x + 2)^3$ with respect to x . We initialize $x = 2$, and perform gradient descent with learning rate $\eta = 0.01$.

Calculate the updated values of x after the first and second iterations. **Round your answers to 2 decimal places.**

Gentle reminder: For automated grading to work correctly:

- For each box, enter only a number.
- The number can include a negative sign (-), digits (0-9), and a decimal point.
- Do not include any letters, spaces, commas, or other characters.

Value of x after the first iteration: _____

Value of x after the second iteration: _____

Ans: The updated value of x after the second iteration is **1.52** (acceptable range: 1.42-1.62)

The updated value of x after the second iteration is **1.15** (acceptable range: 1.05-1.25)